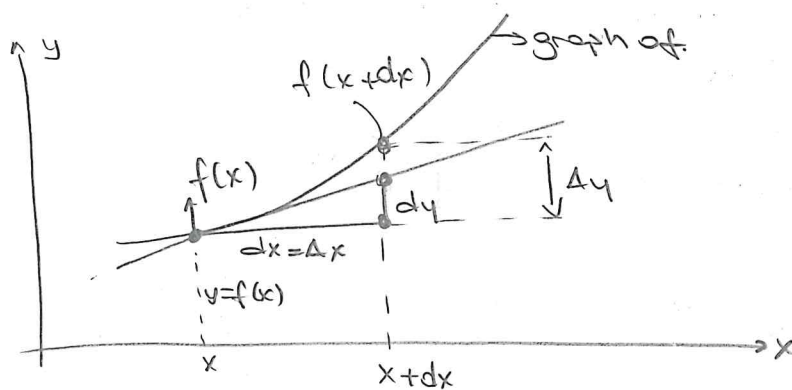


MATH 123 LECTURE NOTES

WEEK (II)

USING DIFFERENTIAL AND DERIVATIVES

Approximating Small Changes



The exact change Δy in y is given by

$$\Delta y = f(x + \Delta x) - f(x)$$

$$\Delta y = \frac{\Delta y}{\Delta x} \Delta x \approx \frac{dy}{dx} \Delta x = f'(x) \Delta x$$

If Δx small $\frac{\Delta y}{\Delta x} \approx \frac{dy}{dx}$

$$\Delta y \approx dy = f'(x) dx$$

Ex 7 Without using a scientific calculator, determine by approximately how much the value of $\sin x$ increases as x increases from $\pi/3$ to $(\pi/3) + 0.006$. To 3 decimal places, what is the value of $\sin(\pi/3 + 0.006)$

If $y = \sin x$ $x = \pi/3 \approx 1.0472$ and $dx = 0.006$

$$dy = \cos x dx = \cos(\pi/3) dx = \frac{1}{2}(0.006) = 0.003$$

Thus the change in the value of $\sin x$ is approximately

0.003 and

$$\underbrace{\sin(\pi/3 + 0.006)}_{f(x+\Delta x)} \approx \underbrace{\sin \pi/3}_{f(x)} + \underbrace{0.003}_{dy \text{ (instead of } \Delta y)} = \frac{\sqrt{3}}{2} + 0.003 = 0.869$$

rounded to 3 decimal places.

Measuring changes with respect to the size of quantities

→ Sometimes changes in a quantity are measured w.r.t. the size of the change which is relative change

$$\text{relative change in } x = \frac{dx}{x}$$

$$\text{percentage change in } x = 100 \cdot \frac{dx}{x}$$

Ex 7 By approximately what percentage does the area of a circle increase if the radius increases by 2%?

$$A = \pi r^2 \quad \Delta A \approx dA = \frac{dA}{dr} dr = 2\pi r dr$$

$$\frac{\Delta A}{A} \approx \frac{dA}{A} = \frac{2\pi r dr}{\pi r^2} = 2 \cdot \frac{dr}{r}$$

If r increases by 2%, then $dr = \frac{2}{100} r$, so.

$$\frac{\Delta A}{A} \approx 2 \cdot \frac{2}{100} = \frac{4}{100} \text{ Thus, } A \text{ increases by approximately } 0.4\%$$

Average and Instantaneous Rates of Change

→ The average rate of change of a function $f(x)$ with respect to x over the interval from a to $a+h$ is

$$\frac{f(a+h) - f(a)}{h}$$

The (instantaneous) rate of change of f with respect to x at $x = a$ is the derivative

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} \quad \text{provided the limit exists.}$$

Ex How fast is area A of a circle increasing with respect to its radius when the radius is 5m ?

The rate of change of the area with respect to the radius is

$$\frac{dA}{dr} = \frac{d}{dr}(\pi r^2) = 2\pi r$$

When $r = 5\text{m}$, the area is changing at the rate $2\pi \times 5 = 10\pi$

This means that a small change Δr in the radius when the radius is 5 would result in a change of $10\pi \Delta r$ in the area of the circle.

$$\Delta A \approx dA = \left. \frac{dA}{dr} \right|_{r=5} \Delta r = 10\pi \Delta r$$

THE MEAN VALUE THEOREM

Thm: Mean value theorem

Suppose that f is continuous on the closed, finite interval $[a, b]$ and that f is differentiable on the open interval (a, b) .

Then, there exists a point c in the open interval

(a, b) such that

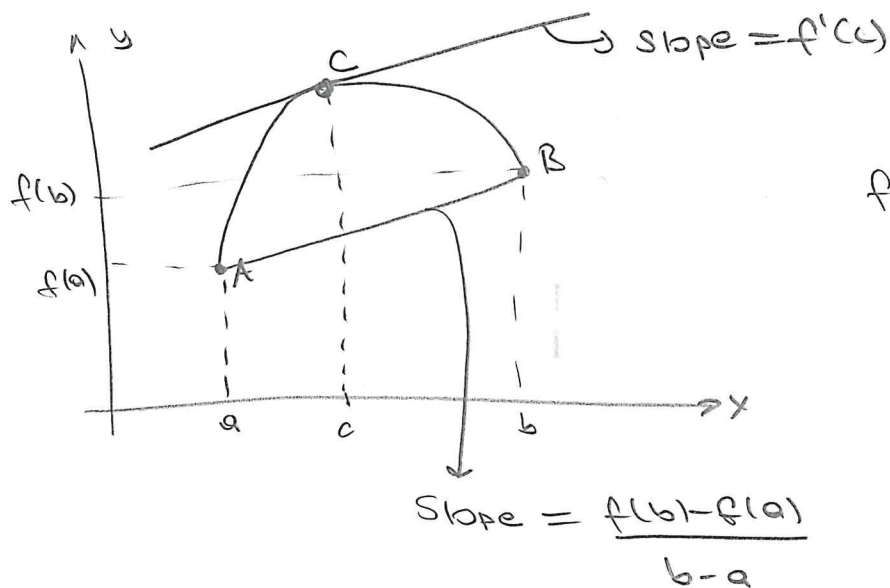
$$\frac{f(b) - f(a)}{b - a} = f'(c)$$

This says that the slope of the chord line joining the points $(a, f(a))$ and $(b, f(b))$ is equal to the slope of the tangent line to the curve $y = f(x)$ at the point $(c, f(c))$ so the two lines are parallel.

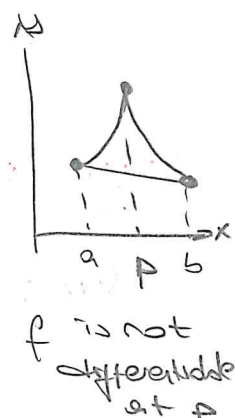
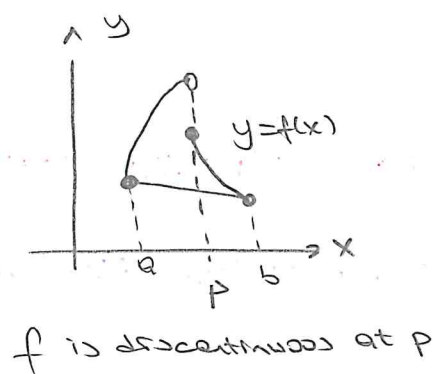
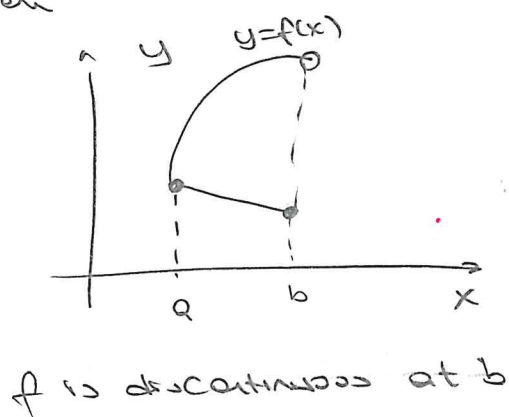
[Chord line: line joining two points on the curve]



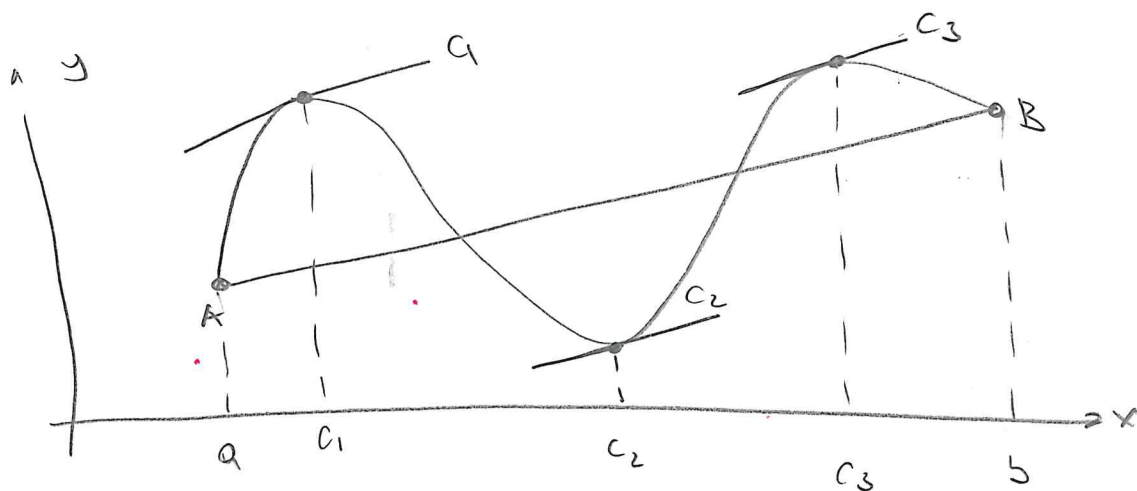
Chord line



→ following functions doesn't satisfy the conditions of the Mean-Value Theorem



NOTE The mean value theorem gives no indication of how many points C there may be on the curve A and B where the tangent is parallel to AB .



Ex 7 Show that $\sin x < x$ for all $x > 0$

→ If $x > 2\pi$, then $\sin x \leq 1 < 2\pi < x$

If $0 < x \leq 2\pi$ then by the mean value theorem

$$\frac{\sin x - \sin 0}{x - 0} = \left. \frac{d}{dx} \sin x \right|_{x=c} = \cos c < 1.$$

Thus, $\sin x < x$

Ex 7 Show that $\sqrt{1+x} < 1 + \frac{x}{2}$ for $x > 0$ and for $-1 < x < 0$

→ If $x > 0$, apply mean value theorem in the interval $[0, x]$. There exists $c \in (0, x)$ such that

$$\frac{\sqrt{1+x} - 1}{x - 0} = \frac{f(x) - f(0)}{x - 0} = f'(c) = \frac{1}{2\sqrt{1+c}} < \frac{1}{2}$$

$c > 0, \sqrt{1+c} > 1$

Then, we have $\sqrt{1+x} < \frac{x}{2} + 1$ ($x > 0$),

→ consider $-1 < x < 0$. Then we have $c \in (x, 0)$

$$\frac{\sqrt{1+x} - 1}{x - 0} = f'(c) = \frac{1}{2\sqrt{1+c}} > \frac{1}{2}$$

$-1 < c < 0$
 $0 < c+1 < 1$
 $\frac{1}{\sqrt{1+c}} > 1$

Then, we get $\sqrt{1+x} - 1 < \frac{x}{2}$ ($x < 0$)

$$\sqrt{1+x} < 1 + \frac{x}{2}$$

Mathematical Consequences

Corollary If $f'(x) = 0$ at each point x of an open interval (a, b) , then $f(x) = C$ for all $x \in (a, b)$ where

C is a constant

$$\left[\frac{f(b) - f(a)}{b - a} = \underbrace{f'(c)}_0 \quad f(b) = f(a) \right] \text{ for all } (a, b)$$

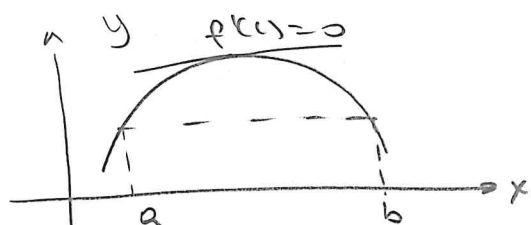
(5)

COROLLARY If $f'(x) = g'(x)$ at each point x in an open interval (a, b) then there exist a constant C such that $f(x) = g(x) + C$ for all $x \in (a, b)$. That is $f - g$ is a constant function on (a, b)

[If $f'(x) = g'(x)$ $(f - g)' = 0$ using previous Corollary we can say $f - g = C$]

THM: ROLLE'S THEOREM

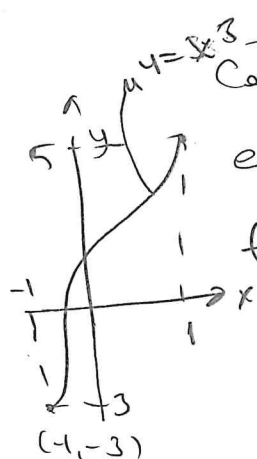
→ Suppose that $y = f(x)$ is continuous at every point of the closed interval $[a, b]$ and differentiable at every point of its interior (a, b) . If $f(a) = f(b)$, then there exists one number c in (a, b) such that $f'(c) = 0$



Result of Mean Value Theorem

$$\frac{f(b) - f(a)}{b - a} = f'(c) = 0$$

Ex 7 Show that the equation $x^3 + 3x + 1 = 0$ has exactly one ^{real} solution



Consider the derivative $f'(x) = 3x^2 + 3$. If there were

even two points such that $x = a$ and $x = b$ where $f(x)$ was zero, Rolle's Theorem would guarantee the existence of a point c such that $f'(c) = 0$. But

$f' \neq 0$ for any x value. The Intermediate Value Theorem

tells us that the graph of $y = f(x)$ has a root in between $(-1, 0)$ $f(-1) = -3 < 0$ $f(0) = 1 > 0$

(6)

THM The Generalized Mean-Value Theorem

If functions f and g are both continuous on $[a, b]$ and differentiable on (a, b) and if $g'(x) \neq 0$ for every $x \in (a, b)$, then there exists a number $c \in (a, b)$ such that

$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(c)}{g'(c)}$$

Proof Note that $g(b) \neq g(a)$; otherwise there would be some number in (a, b) where $g'(c) = 0$. Hence, neither denominator above can be zero. Apply the Mean Value Theorem to

$$h(x) = (f(b) - f(a))(g(x) - g(a)) - (g(b) - g(a))(f(x) - f(a))$$

Since $h(a) = h(b) = 0$, then there exists $c \in (a, b)$ such that $h'(c) = 0$, thus,

$$(f(b) - f(a))g'(c) - (g(b) - g(a))f'(c) = 0$$

Finally, we get

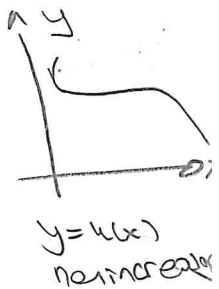
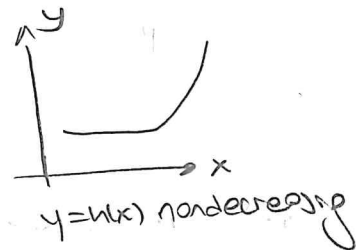
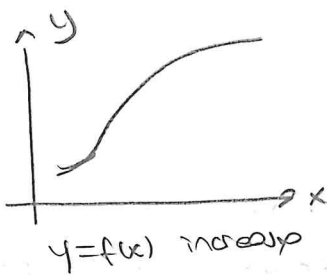
$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(c)}{g'(c)}$$

INCREASING AND DECREASING FUNCTIONS

Definition

Suppose that the function f is defined on an interval I , and that

- (a) If $f(x_2) > f(x_1)$ whenever $x_2 > x_1$, we say f is **increasing** on I .
- (b) If $f(x_2) < f(x_1)$ whenever $x_2 > x_1$, we say f is **decreasing** on I .
- (c) If $f(x_2) \leq f(x_1)$ whenever $x_2 > x_1$, we say f is **nondecreasing** on I .
- (d) If $f(x_2) \geq f(x_1)$ whenever $x_2 > x_1$, we say f is **nonincreasing** on I .



THM Let J be an open interval, and let I be an interval consisting of all points in J and possibly one or both of the endpoints of J . Suppose that f is continuous on I and differentiable on J .

- (a) If $f'(x) > 0$ for all x in J , then f is **increasing** on I .
- (b) If $f'(x) < 0$ for all x in J , then f is **decreasing** on I .
- (c) If $f'(x) \geq 0$ for all x in J , then f is **nondecreasing** on I .
- (d) If $f'(x) \leq 0$ for all x in J , then f is **nonincreasing** on I .

Ex 7 On what intervals is the function $f(x) = x^3 - 12x + 1$ increasing? decreasing?

$$f'(x) = 3x^2 - 12 = 3(x^2 - 4) = 3(x-2)(x+2)$$

$f'(x) > 0$ for $x < -2$ or $x > 2$ increasing

$f'(x) < 0$ for $-2 < x < 2$ decreasing

Ex 7 Show that $f(x) = x^3$ is increasing on any interval

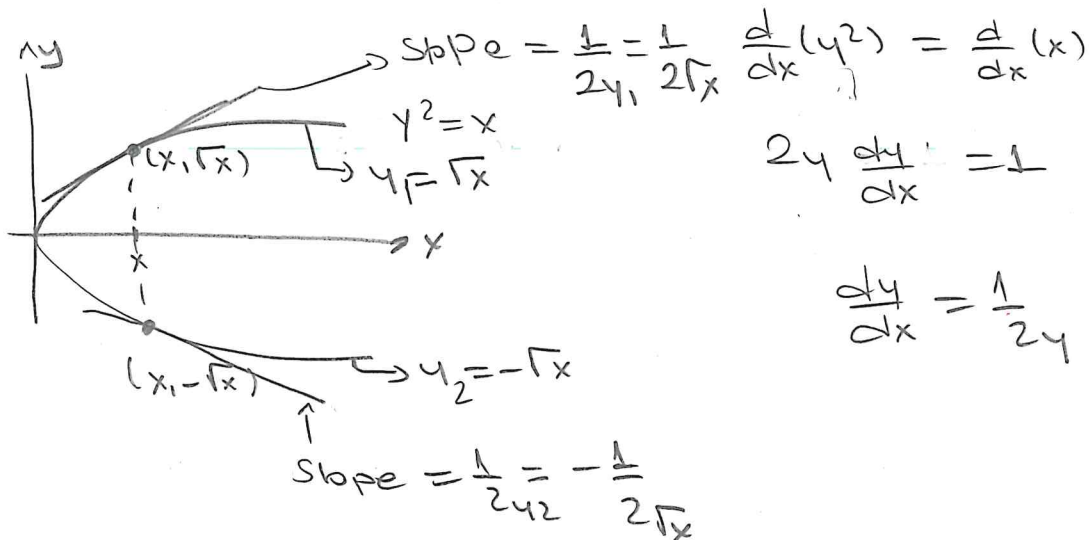
$$f'(x) = 3x^2 > 0 \text{ for all } x \in \mathbb{R}$$

IMPLICIT DIFFERENTIATION

→ Curves are generally the graphs of equations in two variables. Such equations can be written in the form

$F(x, y) = 0 \rightarrow F(x, y)$ denotes an expression involving two variables x and y .

Ex 7 Find $\frac{dy}{dx}$ if $y^2 = x$



$$\frac{d}{dx}(y^2) = \frac{d}{dx}(x)$$

$$2y \frac{dy}{dx} = 1$$

$$\frac{dy}{dx} = \frac{1}{2y}$$

Ex 7 Find slope of the circle $x^2 + y^2 = 25$ at the point $(3, -4)$

$$\frac{d}{dx}(x^2) + \frac{d}{dx}(y^2) = \frac{d}{dx}(25)$$

$$2x + 2y \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{x}{y} \quad \left(\frac{dy}{dx} \right)_{(3, -4)} = -\frac{3}{-4} = \frac{3}{4}$$

Ex₇ Find $\frac{dy}{dx}$ if $y \sin x = x^3 + \cos y$

$$\frac{dy}{dx} \sin x + y \cos x = 3x^2 + \sin y \frac{dy}{dx}$$

$$\frac{dy}{dx} (\sin x + \sin y) = 3x^2 - y \cos x$$

$$\frac{dy}{dx} = \frac{3x^2 - y \cos x}{\sin x + \sin y}$$

Ex₇ Obtain the tangent and normal lines of $x^3 + y^3 - 9xy = 0$ at point (2,4)

$$3x^2 + 3y^2 y' - 9[y + x y'] = 0$$

$$y'(3y^2 - 9x) = 9y - 3x^2$$

$$y' = \frac{9y - 3x^2}{3y^2 - 9x} \bigg|_{(2,4)} = \frac{4}{5} = m_T$$

$$m_N = -\frac{1}{m_T} = -\frac{5}{4}$$

Tangent line $y - 4 = \frac{4}{5}(x - 2)$

Normal line $y - 4 = -\frac{5}{4}(x - 2)$

Ex₇ Obtain $\frac{dy}{dx}$ of the function $y \sin(1/y) = 1 - xy$

$$\frac{dy}{dx} \sin(1/y) + \cos(1/y) (-1/y^2) \frac{dy}{dx} = -y - x \frac{dy}{dx}$$

$$\frac{dy}{dx} [\sin(1/y) - 1/y^2 \cos(1/y) + x] = -y$$

$$\frac{dy}{dx} = \frac{-y}{\sin(1/y) - 1/y^2 \cos(1/y) + x}$$

Ex Obtain $\frac{dy}{dx}$ for the following functions

① $x^2y + xy^2 = 6$

$$2 \cdot x \cdot y + x^2 \frac{dy}{dx} + y^2 + x \cdot 2y \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} [x^2 + 2xy] = -2xy - y^2$$

$$\frac{dy}{dx} = \frac{-2xy - y^2}{x^2 + 2xy}$$

② $x^3 + y^3 = 18xy$

$$3x^2 + 3y^2 \frac{dy}{dx} = 18[y + x \frac{dy}{dx}]$$

$$\frac{dy}{dx} [3y^2 - 18x] = 18y - 3x^2$$

$$\frac{dy}{dx} = \frac{18y - 3x^2}{3y^2 - 18x}$$

③ $xy = \cot(xy)$

$$x \frac{dy}{dx} + y = -\operatorname{cosec}^2(xy) [y + x \frac{dy}{dx}]$$

$$\frac{dy}{dx} [x + x \operatorname{cosec}^2(xy)] = -y - \operatorname{cosec}^2(xy) y$$

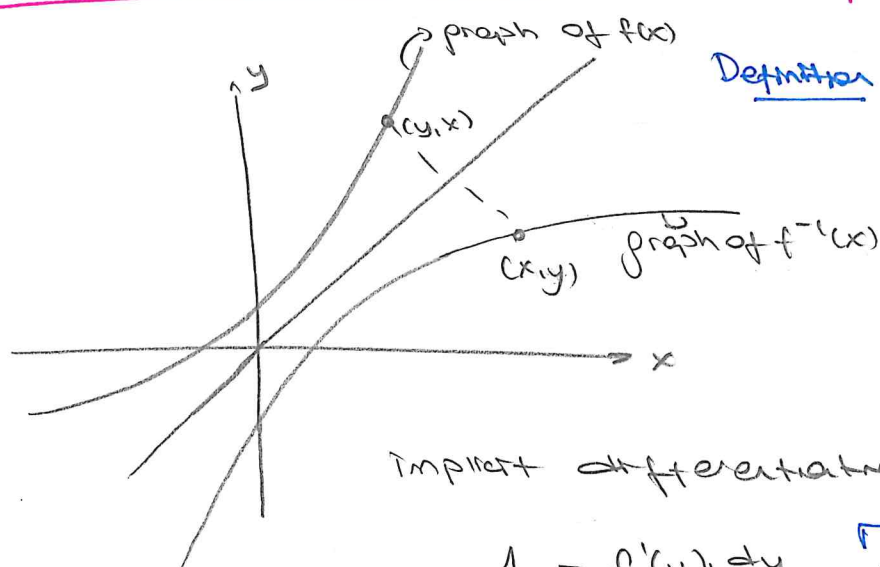
$$\frac{dy}{dx} = - \frac{(y + \operatorname{cosec}^2(xy) \cdot y)}{x(1 + \operatorname{cosec}^2(xy))} = -\frac{y}{x}$$

④ $x + \tan(xy) = 6$

$$1 + \sec^2(xy) [y + x \frac{dy}{dx}] = 0$$

$$\frac{dy}{dx} = - \frac{1 - \sec^2(xy) \cdot y}{\sec^2(xy) \cdot x}$$

DERIVATIVES OF INVERSE FUNCTIONS



Definition

Let $y = f^{-1}(x)$. We want to find dy/dx

Solve the equation

$$y = f^{-1}(x) \text{ for}$$

$$x = f(y) \text{ and use}$$

Implicit differentiation.

$$1 = f'(y) \cdot \frac{dy}{dx}$$

$$\frac{dy}{dx} = \frac{1}{f'(f^{-1}(x))} = \frac{1}{f'(y)}$$

Ex 7 Show that $f(x) = x^3 + x$ is one-to-one on the whole real line and noting that $f(2) = 10$ find $(f^{-1})'(10)$

Since $f'(x) = 3x^2 + 1 > 0$ for all real numbers x , f is increasing and therefore one-to-one and invertible. If $y = f^{-1}(x)$, then

$$x = f(y) = y^3 + y \Rightarrow 1 = (3y^2 + 1) y' \quad y' = \frac{1}{3y^2 + 1}$$

$$f(2) = 10 \quad 2 = f^{-1}(10) = y$$

$$y' = \frac{1}{3y^2 + 1} \Big|_{y=2} = (f^{-1})'(10) = \frac{1}{13}$$

Ex 7 Find $\frac{dy}{dt}$ if $y = (\sin t)^{\ln t}$ $0 < t < \pi$ ($\sin t > 0$)

$$\ln y = \ln t \cdot \ln \sin t$$

$$\frac{y'}{y} = \left(\frac{1}{t} \cdot \ln \sin t + \ln t \cdot \frac{\cos t}{\sin t} \right)$$

$$y' = (\sin t)^{\ln t} \cdot \left(\frac{\ln \sin t}{t} + \ln t \cdot \cot t \right)$$

Ex 7 Differentiate $y = (x+1)(x+2)(x+3)/(x+4)$

$$\ln |y| = \ln |x+1| + \ln |x+2| + \ln |x+3| - \ln |x+4|$$

$$\frac{y'}{y} = \frac{1}{x+1} + \frac{1}{x+2} + \frac{1}{x+3} - \frac{1}{x+4}$$

$$y' = \underbrace{\frac{(x+1)(x+2)(x+3)}{(x+4)}}_y \left(\frac{1}{x+1} + \frac{1}{x+2} + \frac{1}{x+3} - \frac{1}{x+4} \right)$$

Ex 7 Find $\frac{du}{dx}$ if $u = \sqrt{(x+1)(x^2+1)(x^3+1)}$

$$\ln u = \frac{1}{2} [\ln(x+1) + \ln(x^2+1) + \ln(x^3+1)]$$

$$\frac{u'}{u} = \frac{1}{2} \left[\frac{1}{x+1} + \frac{2x}{x^2+1} + \frac{3x^2}{x^3+1} \right]$$

$$u' = \frac{\sqrt{(x+1)(x^2+1)(x^3+1)}}{2} \left[\frac{1}{x+1} + \frac{2x}{x^2+1} + \frac{3x^2}{x^3+1} \right]$$

$$u' \Big|_{x=1} = \frac{\sqrt{8}}{2} \left[\frac{1}{2} + \frac{2}{2} + \frac{3}{2} \right] = 3\sqrt{2}$$

DERIVATIVES OF LOGARITHMIC AND EXPONENTIAL FUNCTIONS

$$\frac{d}{dx} (\log_a x) = \frac{1}{\ln a \cdot x} \quad \text{In particular, } \frac{d}{dx} (\ln x) = \frac{1}{x}$$

$$y = a^x \Rightarrow x = \log_a y$$

$$\Rightarrow 1 = \frac{1}{\ln a \cdot y} \cdot y'$$

$$y = a^x \Rightarrow y' = \ln a \cdot y = \ln a \cdot a^x$$

$$y = (f(x))^{g(x)} \quad (f \text{ or } f(x) > 0) \quad \ln y = g(x) \ln f(x)$$

$$y'/y = (g(x) \cdot \ln f(x))'$$

Ex 7 Find the derivatives of (a) $\ln |\cos x|$ (b) $\ln(x + \sqrt{x^2 + 1})$

$$(a) \frac{d}{dx} \ln |\cos x| = \frac{1}{\cos x} (-\sin x) = -\tan x$$

$$(b) \frac{d}{dx} \ln(x + \sqrt{x^2 + 1}) = \frac{1}{x + \sqrt{x^2 + 1}} \left(1 + \frac{2x}{2\sqrt{x^2 + 1}} \right)$$

$$= \frac{1}{x + \sqrt{x^2 + 1}} + \frac{x}{(x + \sqrt{x^2 + 1})(\sqrt{x^2 + 1})}$$

$$= \frac{x + \sqrt{x^2 + 1}}{(x + \sqrt{x^2 + 1})(\sqrt{x^2 + 1})} = \frac{1}{\sqrt{x^2 + 1}}$$

Ex 7 Find y' if $y = x^x$ for $x > 0$

$$\ln y = \ln x^x \Rightarrow \ln y = x \ln x$$

$$\frac{y'}{y} = (\ln x + x \cdot \frac{1}{x}) \Rightarrow y' = y (\ln x + 1)$$

$$y' = x^x (\ln x + 1)$$

DERIVATIVES OF INVERSE TRIGONOMETRIC FUNCTIONS

→ If $y = \arcsin x$, then $x = \sin y$ and $-\pi/2 \leq y \leq \pi/2$.

Differentiating with respect to x , we obtain:

$$1 = \cos y \frac{dy}{dx} \quad \text{since } -\pi/2 \leq y \leq \pi/2, \cos y \geq 0.$$

Therefore, $\cos y = \sqrt{1 - \sin^2 y} = \sqrt{1 - x^2}$

$$\frac{dy}{dx} = \frac{1}{\cos y} = \frac{1}{\sqrt{1-x^2}}$$

$$\boxed{y = \arcsin x \quad y' = \frac{1}{\sqrt{1-x^2}}}$$

Similarly, we get

****** $\frac{d}{dx} \arccos x = -\frac{1}{\sqrt{1-x^2}}$

$$\frac{d}{dx} \operatorname{arctan} x = \frac{1}{1+x^2}$$

$$\frac{d}{dx} \operatorname{arccot} x = -\frac{1}{1+x^2}$$

$$\frac{d}{dx} \operatorname{arcsec} x = \frac{1}{|x| \sqrt{x^2-1}}$$

$$\frac{d}{dx} \operatorname{arccosec} x = -\frac{1}{|x| \sqrt{x^2-1}}$$

$$\frac{d}{dx} \operatorname{arcsin} x = \frac{1}{\sqrt{1-x^2}}$$

DERIVATIVES OF HYPERBOLIC FUNCTIONS

$$\frac{d}{dx} \cosh x = \sinh x$$

$$\frac{d}{dx} \sinh x = \cosh x$$

$$\frac{d}{dx} \operatorname{sech} x = -\operatorname{sech} x \tanh x$$

$$\frac{d}{dx} \tanh x = \operatorname{sech}^2 x$$

$$\frac{d}{dx} \coth x = -\operatorname{cosech}^2 x$$

$$\frac{d}{dx} \operatorname{cosech} x = -\operatorname{cosech} x \coth x$$

MORE APPLICATIONS OF DIFFERENTIATION

① Related Rates Problems

② Root Finding Problems

③ Evaluation of limits (L'Hopital's)

④ Sketching graphs

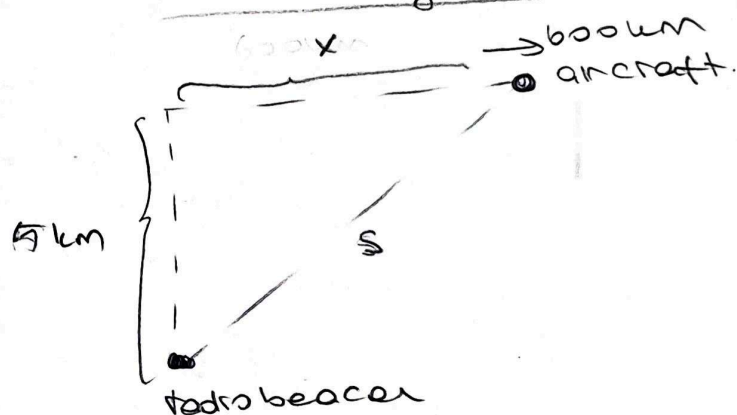
⑤ Extreme value problems (Optimization problems, maximizing and minimizing a quantity)

⑥ Approximation problems (Linear Approximation)

① RELATED RATES PROBLEMS

Ex An aircraft is flying horizontally at a speed of 600 km/h.
How fast is the distance between the aircraft and a radio beacon increasing 1 min after the aircraft passes

5 km directly above the beacon?



$$S^2 = x^2 + 5^2$$

We differentiate this equation implicitly with respect to t to obtain

$$2S \cdot \frac{dS}{dt} = 2x \cdot \frac{dx}{dt}$$

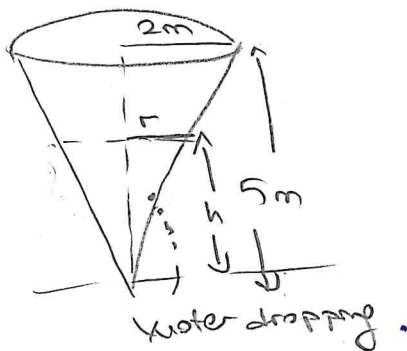
We are given that $\frac{dx}{dt} = 600 \text{ km/h} = 10 \text{ km/min}$

Therefore, $x = 10 \text{ km}$ at time $t = 1$.

$$S^2 = x^2 + 5^2 = 10^2 + 5^2 = 125$$

$$\frac{dS}{dt} = \frac{x}{S} \frac{dx}{dt} = \frac{10}{5\sqrt{5}} \cdot 600 \approx 536.7 \text{ km/h}$$

Ex 7. A leaking water tank is in the shape of an inverted right circular cone with depth 5m and top radius 2m. When the water in the tank is 4m deep, it is leaking out at a rate $\frac{1}{12} \text{ m}^3/\text{min}$. How fast is the water level in the tank dropping at that time?



$$V = \frac{1}{3} \pi r^2 h$$

$$\left. \frac{dV}{dt} \right|_{h=4} = -\frac{1}{12} \text{ m}^3/\text{min}$$

$$\left. \frac{dh}{dt} \right|_{h=4} = ?$$

Similar triangles $\frac{2}{r} = \frac{5}{h} \Rightarrow r = \frac{2h}{5}$ $V = \frac{1}{3} \pi \left(\frac{2h}{5} \right)^2 h$

$$\frac{dV}{dt} = \frac{1}{3} \pi \left[\frac{4}{25} \cdot 3h^2 \frac{dh}{dt} \right]_{h=4} = \frac{4\pi}{25} \cdot 16 \cdot \left. \frac{dh}{dt} \right|_{h=4} = -\frac{1}{12}$$

$$\left. \frac{dh}{dt} \right|_{h=4} = -\frac{25}{768\pi}$$

FINDING ROOTS OF EQUATIONS

FIXED POINT ITERATION

→ Looking for solutions of equations of the form

$$f(x) = x$$

called fixed points of f

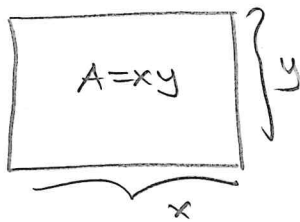
→ Starting from a number x_0 , we generate a

sequence $x_1, x_2, \dots$ of numbers by the rule

$$x_{n+1} = f(x_n)$$

In certain circumstances, $\lim x_n$ exists and converges to a fixed point.

Ex How fast is the area of a rectangle changing if one side is 10cm long and is increasing at a rate 2cm/s and the other side is 8cm long and decreasing at a rate 3cm/s



$$x = 10$$

$$\frac{dx}{dt} = 2$$

$$\frac{dA}{dt} = ?$$

$$y = 8$$

$$\frac{dy}{dt} = -3$$

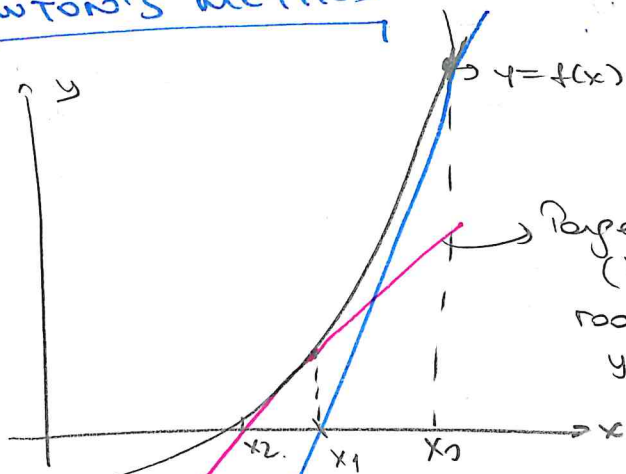
Decreasing -
Increasing +

$$\frac{dA}{dt} = x \cdot \frac{dy}{dt} + y \cdot \frac{dx}{dt} = 10(-3) + 8 \cdot 2 = -64$$

Procedures for Related-Rates Problems

1. Read the problem very carefully. Try to understand the relationship between the variable quantities. What is given? What is to be found?
2. Make a sketch if appropriate
3. Define variables and express given, required quantities in terms of these variables.
4. Identify the relations between variables
5. Differentiate variables based on the given independent variable (For example take derivatives w.r.t time if variables are function of time)
6. Substitute given values, then solve the equations

NEWTON'S METHOD



Looking for fixed point of the function

$$g(x) = x - \frac{f(x)}{f'(x)}$$

root of $y = f(x_0) + f'(x_0)(x - x_0)$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

Tangent line at $(x_0, f(x_0))$

$$y = f(x_0) + f'(x_0)(x - x_0)$$

Iteration formula $x_{n+1} = \frac{x_n}{f'(x_n)} - \frac{f(x_n)}{f'(x_n)}$

Ex 7 Use Newton's method to find the root $x^3 - x - 1 = 0$

$$f(x) = x^3 - x - 1 \quad f'(x) = 3x^2 - 1$$

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = x_n - \frac{x_n^3 - x_n - 1}{3x_n^2 - 1}$$

<u>n</u>	<u>x_n</u>	<u>$f(x_n)$</u>
0	1.5	0.87500 - - -
1	1.347	0.1
2	1.32571	0.002
3	1.324718	0.000000924
4	1.324717	0.000000 - - - 0

THM A fixed-point theorem

Suppose that f is defined on an interval $I = [a, b]$ and satisfies the following two conditions:

- (i) $f(x)$ belongs to I whenever x belongs to I and
- (ii) there exists a constant k with $0 < k < 1$ such that for every u and v in I ,

$$|f(u) - f(v)| \leq k|u - v|$$

Then, f has a unique fixed point r in I , that is,

$f(r) = r$, and starting with any number x_0 in I , the iterates

$$x_1 = f(x_0), x_2 = f(x_1), \dots \text{ converge to } r.$$

Ex 7 Find a root of the equation $\cos x = 5x$.

This equation is of the form $f(x) = x$ and $f(x) = \frac{1}{5} \cos x$.

So, we have iteration procedure $x_{k+1} = f(x_k)$

$$x_{k+1} = \frac{1}{5} \cos x_k$$

→ To initiate, we must find a reasonable x_0 . Since $\cos x$ is close to 1 for x near 0, we see that $\frac{1}{5} \cos x$ will be close to $\frac{1}{5}$ when $x = \frac{1}{5}$. Therefore, choose $x_0 = \frac{1}{5}$.

$$x_1 = \frac{1}{5} \cos x_0, x_2 = \frac{1}{5} \cos x_1, x_3 = \frac{1}{5} \cos x_2$$

The root is 0.19616428 to 8 decimal

n	x_n
0	0.2
1	0.1960133
2	0.1961701
3	0.1961641
4	0.1961641
5	0.1961641
6	0.1961641

INDETERMINATE FORMS

Type

0/0

Example

$$\lim_{x \rightarrow 0} \frac{\sin x}{x}$$

∞/∞

$$\lim_{x \rightarrow 0} \frac{\ln(1/x^2)}{\cot(x^2)}$$

0 $\cdot$ ∞

$$\lim_{x \rightarrow 0^+} x \ln 1/x$$

$\infty - \infty$

$$\lim_{x \rightarrow (\pi/2)} \left(\tan x - \frac{1}{\pi - 2x} \right)$$

0⁰

$$\lim_{x \rightarrow 0^+} x^x$$

∞^0

$$\lim_{x \rightarrow (\pi/2)} (\tan x)^{\cos x}$$

1 ^{∞}

$$\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x} \right)^x$$

THM The first L'Hospital Rule

Suppose that functions f and g are differentiable on the interval (a, b) and $g'(x) \neq 0$ there. Suppose also that

(i) $\lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow a^+} g(x) = 0$ and

(ii) $\lim_{x \rightarrow a^+} \frac{f'(x)}{g'(x)} = L$ (where L is finite or ∞ or $-\infty$)

Then $\lim_{x \rightarrow a^+} \frac{f(x)}{g(x)} = L$

Similar results hold if everywhere of $\lim_{x \rightarrow a^+}$ is replaced by $\lim_{x \rightarrow b^-}$ or even $\lim_{x \rightarrow c}$ where $a < c < b$

The cases $a = -\infty$ and $b = \infty$ are also involved.

Ex 7 $\lim_{x \rightarrow 1} \frac{\ln x}{x^2 - 1} = ?$

$$\lim_{x \rightarrow 1} \frac{\ln x}{x^2 - 1} = \frac{0}{0}$$

$$\lim_{x \rightarrow 1} \frac{1/x}{2x} = \lim_{x \rightarrow 1} \frac{1}{2x^2} = \frac{1}{2}$$

Ex 7 $\lim_{x \rightarrow 0} \frac{2 \sin x - \sin(2x)}{2e^x - 2 - 2x - x^2} = ? \quad \left(\frac{0}{0}\right)$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{2 \sin x - \sin 2x}{2e^x - 2 - 2x - x^2} = \lim_{x \rightarrow 0} \frac{2 \cos x - 2 \cos(2x)}{2e^x - 2 - 2x}$$

$$= \lim_{x \rightarrow 0} \frac{\cos x - \cos(2x)}{e^x - 1 - x} \left(\frac{0}{0}\right) = \lim_{x \rightarrow 0} \frac{-\sin x + \sin(2x) \cdot 2}{e^x - 1} \left(\frac{0}{0}\right)$$

$$= \lim_{x \rightarrow 0} \frac{-\cos x + \cos(2x) \cdot 4}{e^x} = \frac{3}{1} = 3$$

Ex 7 Evaluate (a) $\lim_{x \rightarrow (\pi/2)^-} \frac{2x - \pi}{\cos^2 x}$ (b) $\lim_{x \rightarrow 1^+} \frac{x}{\ln x}$

(a) $\lim_{x \rightarrow \pi/2^-} \frac{2x - \pi}{\cos^2 x} = \lim_{x \rightarrow \pi/2^-} \frac{2}{2 \cos x (-\sin x)} = -\infty$

(b) L'Hopital's Rule can't be used to evaluate $\lim_{x \rightarrow 1^+} x / \ln x$ because this is not an indeterminate form. The $\lim_{x \rightarrow 1^+} x = 1$ and $\lim_{x \rightarrow 1^+} \ln x \rightarrow 0$.

$$\lim_{x \rightarrow 1^+} \frac{x}{\ln x} = \infty$$

(If you try L'Hopital rule you will get $\lim_{x \rightarrow 1^+} \frac{1}{1/x} = 1$ wrong!!)

THM The Second L'Hôpital Rule

Suppose that f and g are differentiable on the interval (a, b) and that $g'(x) \neq 0$ there.

Suppose also that

(i) $\lim_{x \rightarrow a^+} g(x) = \pm \infty$ and

(ii) $\lim_{x \rightarrow a^+} \frac{f'(x)}{g'(x)} = L$ (where L is finite or ∞ or $-\infty$)

Then, $\lim_{x \rightarrow a^+} \frac{f(x)}{g(x)} = L$

Again similar results hold for $\lim_{x \rightarrow b^-}$ and for $\lim_{x \rightarrow c}$ and the cases $a = -\infty$ and $b = \infty$ are allowed

Ex 7 Evaluate (a) $\lim_{x \rightarrow \infty} \frac{x^2}{e^x}$ and (b) $\lim_{x \rightarrow 0^+} x^a \ln x$ where $a > 0$

Solution (a) $\lim_{x \rightarrow \infty} \frac{x^2}{e^x} \stackrel{(\infty/\infty)}{=} \lim_{x \rightarrow \infty} \frac{2x}{e^x} \stackrel{(\infty/\infty)}{=} \lim_{x \rightarrow \infty} \frac{2}{e^x} = 0$

Similarly, one can show that $\lim_{x \rightarrow \infty} \frac{x^n}{e^x} = 0$ for any positive integer n by repeated applications of L'Hôpital Rule

(b) $\lim_{x \rightarrow 0^+} x^a \ln x$ ($a > 0$) $(0, -\infty)$

$$= \lim_{x \rightarrow 0^+} \frac{\ln x}{x^{-a}} \stackrel{(-\infty/\infty)}{=} \lim_{x \rightarrow 0^+} \frac{1/x}{-a \cdot x^{-a-1}} = \lim_{x \rightarrow 0^+} \frac{x^a}{-a} = 0$$

Ex 7

Evaluate

$$\lim_{x \rightarrow 0^+} x^x$$

(0⁰)

$$y = x^x$$

$$\ln y = x \ln x$$

$$\lim_{x \rightarrow 0^+} \ln y = \lim_{x \rightarrow 0^+} x \cdot \ln x$$

$$(0, -\infty) = \lim_{x \rightarrow 0^+} \frac{\ln x}{1/x} \stackrel{(-\infty/\infty)}{=} \lim_{x \rightarrow 0^+} \frac{1/x}{-1/x^2}$$

$$\lim_{x \rightarrow 0^+} \ln y = \lim_{x \rightarrow 0^+} x^x = 0 \quad = 0$$

$$\lim_{x \rightarrow 0^+} x^x = 1$$

$$\lim y = e^{\ln(\lim y)} = e^{\lim(\ln y)}$$

Ex 7

$$\lim_{x \rightarrow \infty} \left(1 + \sin \frac{3}{x}\right)^x = ?$$

$$\lim_{x \rightarrow \infty} \left(1 + \sin \frac{3}{x}\right)^x \stackrel{1^\infty}{=} ?$$

$$y = \left(1 + \sin \frac{3}{x}\right)^x \quad \ln y = x \ln \left(1 + \sin \frac{3}{x}\right)$$

$$\lim_{x \rightarrow \infty} \ln y = \lim_{x \rightarrow \infty} x \ln \left(1 + \sin \frac{3}{x}\right) \stackrel{(0,0)}{=} \lim_{x \rightarrow \infty} \frac{\ln \left(1 + \sin \frac{3}{x}\right)}{1/x}$$

$$= \lim_{x \rightarrow \infty} \frac{\frac{1}{1 + \sin \frac{3}{x}} \cos \left(\frac{3}{x}\right) \left(-\frac{3}{x^2}\right)}{-\frac{1}{x^2}} = \lim_{x \rightarrow \infty} \frac{3 \cos \left(\frac{3}{x}\right) \frac{1}{1 + \sin \frac{3}{x}}}{1}$$

$$= 3$$

$$\lim_{x \rightarrow \infty} \ln \left(1 + \sin \frac{3}{x}\right)^x = 3$$

$$\lim_{x \rightarrow \infty} \left(1 + \sin \frac{3}{x}\right)^x = e^3$$

(23)

Examples Solve following limits

$$(a) \lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} \stackrel{L'H}{=} \lim_{x \rightarrow 0} \frac{\frac{1}{1+x}}{1} = \lim_{x \rightarrow 0} \frac{1}{1+x} = 1$$

$$(b) \lim_{x \rightarrow 0} \frac{e^x - 1 - x}{x^2} \stackrel{L'H}{=} \lim_{x \rightarrow 0} \frac{e^x - 1}{2x} \stackrel{L'H}{=} \lim_{x \rightarrow 0} \frac{e^x}{2} = \lim_{x \rightarrow 0} \frac{e^x}{2} = \frac{1}{2}$$

$\lim_{x \rightarrow 0^-} \frac{e^x}{2} = -\infty$
 $\lim_{x \rightarrow 0^+} \frac{e^x}{2} = \infty$

LIMIT DOES NOT EXIST.

$$(c) \lim_{x \rightarrow \pi/2} \frac{1 - \cos(\cos x)}{\cos x} \stackrel{(0/0)}{=} \lim_{x \rightarrow \pi/2} \frac{\sin(\cos x) (-\sin x)}{(-\sin x)} = 0$$

$$(d) \lim_{x \rightarrow 0} \frac{\sqrt{1+x} - \frac{x}{2} - 1}{x^2} \stackrel{(0/0)}{=} \lim_{x \rightarrow 0} \frac{\frac{1}{2\sqrt{1+x}} - \frac{1}{2}}{2x} = \lim_{x \rightarrow 0} \frac{1}{4} \frac{1 - \sqrt{1+x}}{x\sqrt{1+x}} \stackrel{(0/0)}{=}$$

$$= \lim_{x \rightarrow 0} \frac{1}{4} \left(-\frac{1}{2\sqrt{1+x}} \right) = \lim_{x \rightarrow 0} \frac{1}{4} \left(-\frac{1}{2} \right) = -\frac{1}{8}$$

$$(e) \lim_{x \rightarrow \infty} (x \sin(1/x)) \stackrel{(0 \cdot \infty)}{=} \lim_{x \rightarrow \infty} \frac{\sin(1/x)}{1/x} = \lim_{y=1/x \rightarrow 0} \frac{\sin y}{y} = 1$$

$$(f) \lim_{x \rightarrow 0} \left(\frac{1}{\sin x} - \frac{1}{x} \right) = ? \quad (\infty - \infty)$$

$$\lim_{x \rightarrow 0} \left(\frac{1}{\sin x} - \frac{1}{x} \right) = \lim_{x \rightarrow 0} \frac{x - \sin x}{x \sin x} \stackrel{(0/0)}{=} \lim_{x \rightarrow 0} \frac{1 - \cos x}{\sin x + x \cos x} \stackrel{(0/0)}{=}$$

$$= \lim_{x \rightarrow 0} \frac{\sin x}{\cos x + \cos x - x \sin x} = 0$$

9

$$\lim_{x \rightarrow 0^+} (1+x)^{1/x} \quad (1^\infty)$$

$$\lim_{x \rightarrow 0^+} (1+x)^{1/x} = \lim_{x \rightarrow 0^+} e^{\ln(1+x)^{1/x}} = \lim_{x \rightarrow 0^+} e^{\frac{1}{x} \ln(1+x)}$$

$$= e^{\lim_{x \rightarrow 0^+} \frac{1}{x} \ln(1+x)} = e^1$$

$$\lim_{x \rightarrow 0^+} \frac{1}{x} \ln(1+x) = \lim_{x \rightarrow 0^+} \frac{\frac{1}{1+x}}{\frac{1}{1+x}} = \frac{1}{1} = 1$$

h

$$\lim_{x \rightarrow \pi/2^-} (1-\sin x)^{\cot x} \quad (0^0) = \lim_{x \rightarrow \pi/2^-} e^{\ln(1-\sin x)^{\cot x}}$$

$$= \lim_{x \rightarrow \pi/2^-} e^{\cot x \ln(1-\sin x)} = e^{\lim_{x \rightarrow \pi/2^-} \cot x \ln(1-\sin x)}$$

$$= e^0 = 1$$

$$\lim_{x \rightarrow \pi/2^-} \cot x \ln(1-\sin x) = \lim_{x \rightarrow \pi/2^-} \frac{\ln(1-\sin x)}{\tan x} \quad \left(\frac{0}{0} \right)$$

$$= \lim_{x \rightarrow \pi/2^-} \frac{-\cos x}{1-\sin x} = \lim_{x \rightarrow \pi/2^-} \frac{\cos^3 x}{\sin x - 1} \quad \left(\frac{0}{0} \right)$$

$$= \lim_{x \rightarrow \pi/2^-} \frac{3 \cos^2 x (-\sin x)}{\cos x} = \lim_{x \rightarrow \pi/2^-} 3 \cos x (-\sin x) = 0$$